

Through Their Eyes: A First-Person VR Experience of Alzheimer's Disease

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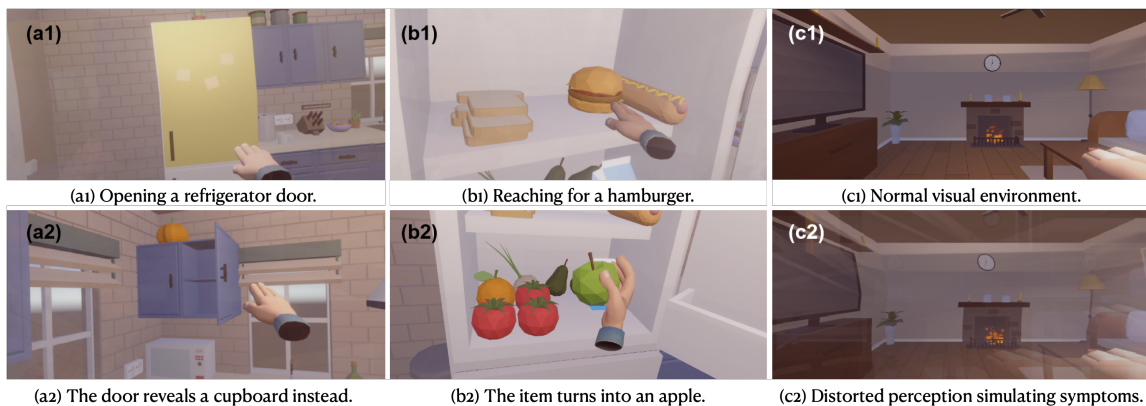


Figure 1: Overview of the in-game simulation for Alzheimer's-related cognitive disorientation.

Abstract

Alzheimer's disease is a common neurodegenerative disorder characterized by symptoms such as memory decline, cognitive disorientation, and perceptual impairments. However, these symptoms are often difficult for the public to intuitively understand, leading to a lack of awareness and empathy for patients' experiences. This study presents an immersive virtual reality (VR) simulation system that

enables users to experience, from a first-person perspective, the daily challenges patients face, thereby enhancing their understanding of the disease. A mixed-methods user study combining questionnaires and interviews showed that the system effectively improved users' empathy and cognitive understanding of Alzheimer's-related conditions.

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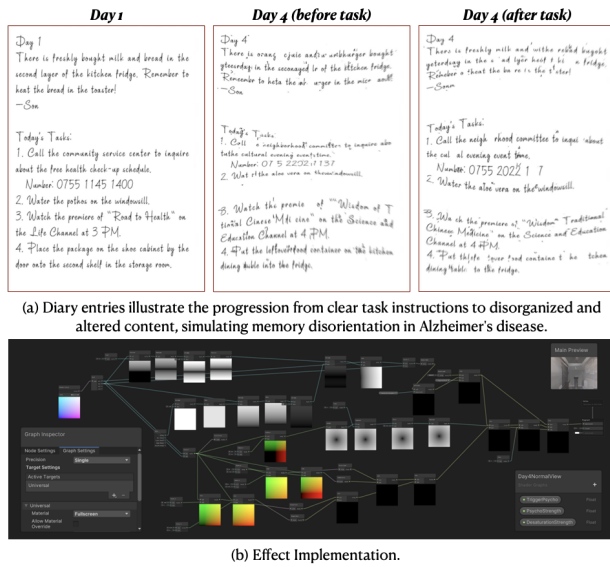


Figure 2: Overview of the workflow.

1 Introduction

In today's society, people with Alzheimer's disease face not only the challenges of the disease itself but also the dual pressures of insufficient understanding and prejudice from caregivers and the public. Without direct experience, it is difficult for the general public to truly perceive the daily struggles caused by memory decline and cognitive impairment, which hinders empathy and the provision of support. This study utilizes the immersion and interactivity of virtual reality (VR) to place participants in a first-person perspective, enabling them to experience patients' living conditions more intuitively. In this project, we aim to enhance public understanding and empathy toward Alzheimer's disease and to stimulate willingness to provide social support (figure 1).

2 Related Works

Recent years have witnessed a growing body of research employing VR and serious game-based (VRGS-based) instruments for evaluating spatial and memory abilities in Alzheimer's disease patients and at-risk populations, showing significant clinical potential for identifying early symptoms of cognitive impairment [Sánchez-Escudero et al. 2025]. Research also indicates that VR-based cognitive training may maintain or improve memory, attention, and spatial navigation, fostering independence and quality of life [Park et al. 2019]. Furthermore, experimental evidence indicates that explorable 360° video can improve informal caregivers' understanding of Alzheimer's symptoms, thereby enhancing their empathy to make their care procedures more effective [Festari et al. 2024].

3 Project Overview

The 3D model "Cartoon House Interiors" [JustCreate 2024] was adapted to create a virtual domestic environment that reflects the daily life of individuals with Alzheimer's disease. The simulation incorporates routine activities, resting, cooking, laundry, dining,

and stair use, to replicate spatial and functional demands. A low-poly, cartoon-style aesthetic, informed by user-centered design, was adopted to reduce cognitive load and aid rapid orientation. This approach enhances immersion and effectively conveys the everyday challenges of cognitive decline.

Interaction Design: In this project, a two-day experiential program simulates the daily challenges of Alzheimer's disease. On "Day 1" (Figure 2a: Day 1), users complete routine tasks as usual to establish their baseline. On "Day 4", the simulation introduces cognitive and perceptual impairments characteristic of Alzheimer's disease through deliberately imposed task obstacles and environmental modifications. Users experience characteristic symptoms while performing daily activities: blurred and distorted vision simulates visuospatial deficits; memos with fragmented text mimic reading difficulties (Figure 2a: Day 4 before and after task). Additionally, experience such as opening the kitchen door then finding it leading to the bathroom (Figure 1a1, a2) or following written instructions (Figure 2) to retrieve orange juice and a hamburger but only to discover the absence of orange juice and the hamburger transforming into an apple upon interaction (Figure 1b1, b2), effectively simulating memory distortions and cognitive disorganization.

Technical Implementation: Our project was developed using Unity 2022.3.5f1c1, with the Universal Render Pipeline (URP) for graphics rendering. The visual effects were created in Shader Graph, employing a center-mirrored mask to achieve edge blurring, applying horizontal distortions to different regions of the screen to create localized displacements, and combining color inversion with dynamic desaturation to produce chromatic abnormalities (Figure 2b). These, along with dynamically controlled parameters, simulate the disoriented visual perception experienced by patients with Alzheimer's disease. Interactive elements were implemented using the XR Interaction Toolkit, in conjunction with Timeline, Animator, and custom scripts, to create scenarios such as consuming the wrong items or entering the wrong room.

4 User Study

We conducted a user study comprising a pre-post three-dimensional Likert-scale questionnaire (knowledge, empathy, attitudes, and behaviors) toward Alzheimer's disease patients and a post-experience semi-structured interview. Six participants (three male, three female) completed the full study procedure. Results showed significant improvements in knowledge (3.04→4.04), empathy (2.92→4.17), and attitudes/behaviors (3.29→4.25). Interviews confirmed that the immersive, first-person format, with simulated symptoms such as tinnitus and visual distortions, was more impactful than videos or text, enhancing willingness to support this group. Participants also suggested improvements, such as adding more interactive characters and expanding scene content.

5 Conclusion and Futurework

This VR project simulates a first-person experience of Alzheimer's disease through abnormal visual, auditory, and behavioral interactions. The user study indicates that the project improves participants' understanding and empathy toward individuals with Alzheimer's disease, suggesting its potential to raise awareness

and encourage real-life support. The current version includes interactive content for “Day 1” (normal state) and “Day 4” (severe symptoms). In the future, we plan to add the remaining stages to illustrate the gradual progression of symptoms, introduce additional interactive characters, scenes, and activities to further enrich the content, and enhance immersion.

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